COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.* (BL4)

Assessment Tool Number : CO2_AT2
Assessment Tool Weightage : 35%
Assessment Tool : Simulation Task
Duration : Self Learning
Marks : 100 Marks
Type of Assessment Conduction : At Home
Evaluation Type : Online Evaluation(GitHub)
Result Generation : Students successfully simulate the assigned wireless communication system, analyze the obtained results, and interpret the performance based on relevant technical parameters.

Topic Covered: LTE/LTE-Advanced architecture, protocol stack, carrier aggregation, higher-order MIMO, beamforming, and beamsteering techniques.
5G NR air interface, gNodeB and RAN architecture, RRC procedures, 5G Core, Service-Based Architecture, network slicing, AMF, SMF, UPF, and control/user plane separation.

Conduction of Assessment Procedure:

- Assessment will be carried out through rubric-based evaluation, where students complete the assigned task.
- Simulation will be done in MATLAB/Simulink for the given specification and interpret the results.
- TWO questions will be assigned to students each carrying 10 marks and they are required to collaboratively discuss the given problem and present their responses with clear justification and submit the document in GitHub Platform.

1.Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)

2.Simulation of 5G core traffic routing under user density, mobility, latency, throughput, and resource utilization (BL4)

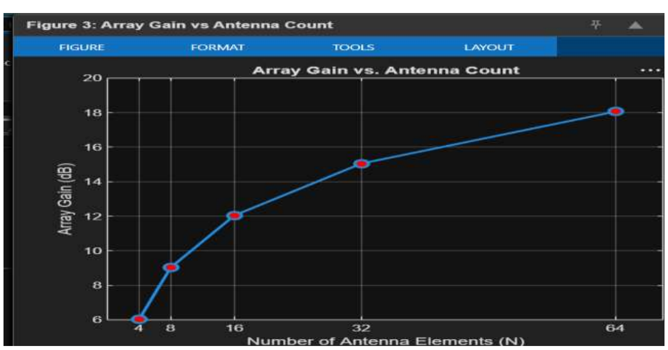
Question 01:

Program:

```
clc;
clear;
close all;
%% 1. Parameters Setup
c = 3e8; % Speed of light (m/s)
fc = 28e9; % Carrier frequency (28 GHz - 5G mmWave)
lambda = c / fc; % Wavelength
d = lambda / 2; % Half-wavelength antenna spacing
N = 8; % Default number of antenna elements
theta = -90:0.1:90; % Angular range in degrees
theta_rad = deg2rad(theta);
%% 2. Analog Beamforming Pattern for Different Steering Angles
beamAngles = [-30, 0, 30]; % Target beam directions in degrees
figure('Name', 'Analog Beamforming Steering Angles');
for k = 1:length(beamAngles)
theta0 = beamAngles(k);
theta0_rad = deg2rad(theta0);
% Progressive phase shift required for beam steering at theta0
beta = -2 * pi * (d / lambda) * sin(theta0_rad);
% Array Factor calculation
psi = 2 * pi * (d / lambda) * sin(theta_rad) + beta;
AF = abs(sin(N * psi / 2) ./ (N * sin(psi / 2)));
AF(isnan(AF)) = 1; % Handle 0/0 limit (main lobe peak)
% Plot normalized radiation pattern in dB
subplot(length(beamAngles), 1, k);
plot(theta, 20*log10(AF), 'LineWidth', 1.8);
grid on;
ylim([-40, 2]);
xlim([-90, 90]);
xlabel('Angle (Degrees)');
ylabel('Normalized Gain (dB)');
title(['Beam Steering Angle (\theta_0) = ', num2str(theta0), '°']);
end
%% 3. Sidelobe Level (SLL) Analysis & Phase Shift Variation Effect
% Simulating phase error/variation (e.g., non-ideal phase shifters)
phase_error_std = 10; % Standard deviation of phase error in degrees
% Ideal steering at 0 degrees
beta_ideal = zeros(1, N);
% Non-ideal steering with random phase shift variation
rng(1); % For reproducibility
phase_noise = deg2rad(normrnd(0, phase_error_std, [1, N]));
beta_noisy = beta_ideal + phase_noise;
% Calculating Array Factor with phase shift variations
AF_ideal = zeros(size(theta));
```

```
AF_noisy = zeros(size(theta));
for i = 1:length(theta)
% Steering vector for element positions [0, 1, ..., N-1]
n = 0:N-1;
spatial_phase = 2 * pi * (d / lambda) * n * sin(theta_rad(i));
AF_ideal(i) = abs(sum(exp(1i * (spatial_phase + beta_ideal))));
AF_noisy(i) = abs(sum(exp(1i * (spatial_phase + beta_noisy))));
end
AF_ideal = AF_ideal / max(AF_ideal);
AF_noisy = AF_noisy / max(AF_noisy);
figure('Name', 'Phase Shift Variation & Sidelobe Level');
plot(theta, 20*log10(AF_ideal), 'b', 'LineWidth', 1.8, 'DisplayName', 'Ideal Phase');
hold on;
plot(theta, 20*log10(AF_noisy), 'r--', 'LineWidth', 1.5, 'DisplayName', 'Phase Error
(\sigma = 10°)');
grid on;
ylim([-40, 2]);
xlim([-90, 90]);
xlabel('Angle (Degrees)');
ylabel('Normalized Gain (dB)');
title('Impact of Phase Shift Variation on Sidelobe Level (SLL)');
legend('Location', 'northeast');
%% 4. Array Gain vs. Antenna Count (N)
antenna_counts = [4, 8, 16, 32, 64];
array_gain_dB = 10 * log10(antenna_counts); % Gain G = 10*log10(N)
figure('Name', 'Array Gain vs Antenna Count');
plot(antenna_counts, array_gain_dB, '-o', 'LineWidth', 2, 'MarkerSize', 8,
'MarkerFaceColor', 'r');
grid on;
xlabel('Number of Antenna Elements (N)');
ylabel('Array Gain (dB)');
title('Array Gain vs. Antenna Count');
xticks(antenna_counts);
```

Outputs:



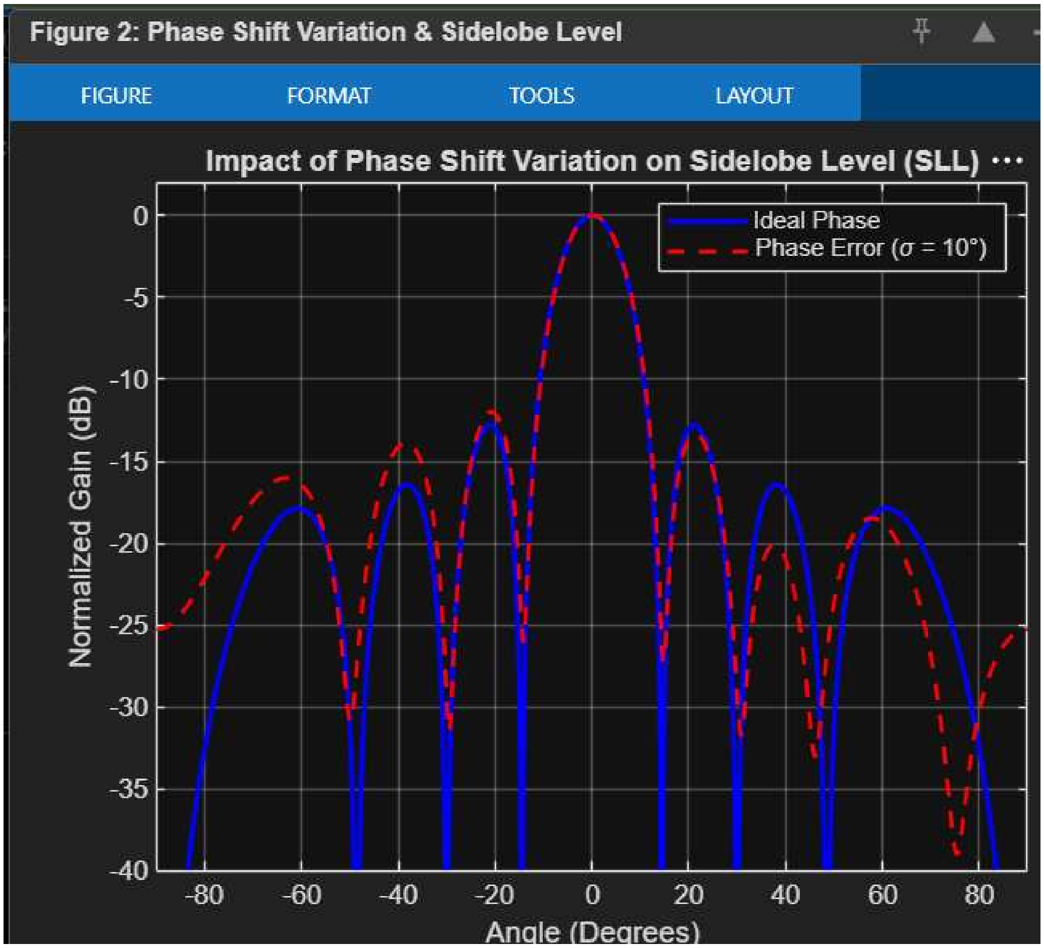
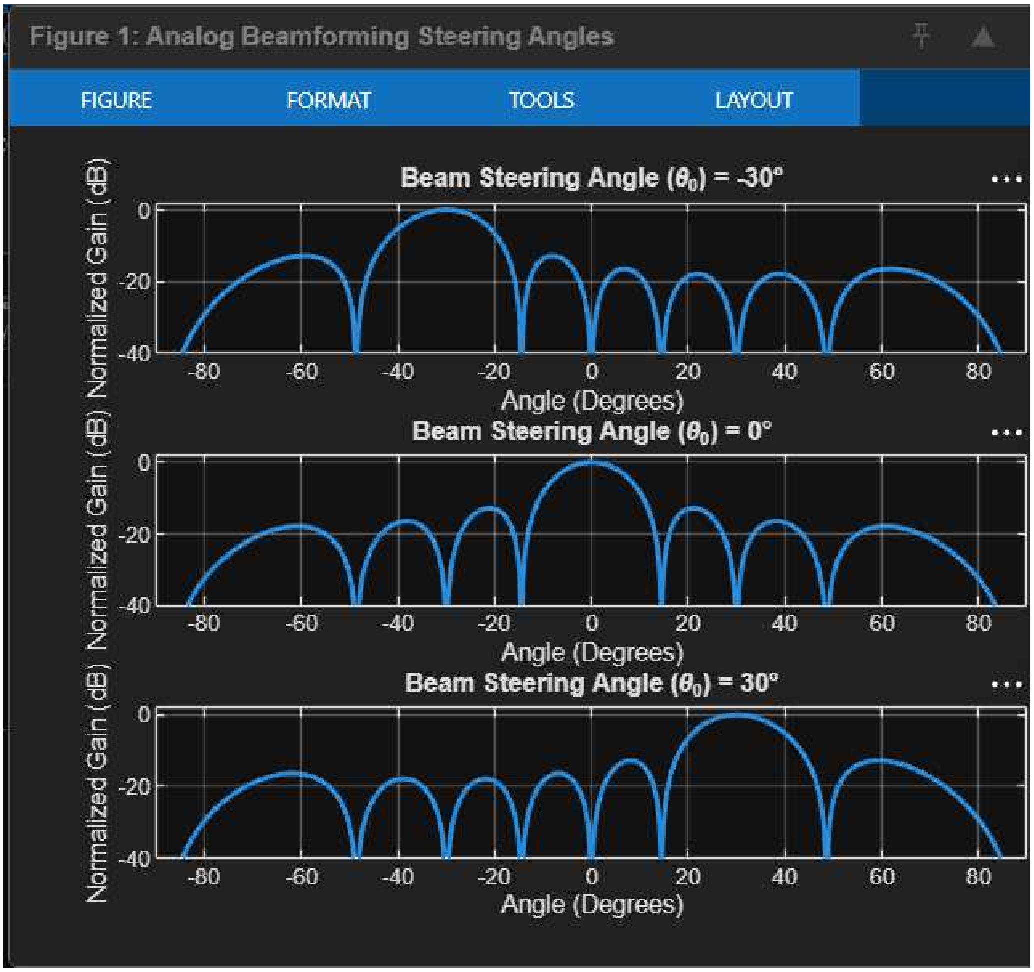


Figure 2: Phase Shift Variation & Sidelobe Level
FIGURE
FORMAT
TOOLS
LAYOUT
Impact of Phase Shift Variation on Sidelobe Level (SLL)
Ideal Phase
Phase Error (σ = 10°)
Normalized Gain (dB)
0
-5
-10
-15
-20
-25
-30
-35
-40
-80
-60
-40
-20
0
20
40
60
80
Angle (Degrees)



Question 2:

Program :

```
clc;
clear;
close all;
%% 5G Core Traffic Routing Simulation
% User Density
userDensity = 100:100:1000;
% Mobility (km/h)
mobility = [10 20 40 60 80 100 120 140 160 180];
% Performance Parameters
latency = 5 + 0.015*userDensity; % ms
throughput = 1200 - 0.6*userDensity; % Mbps
resourceUtilization = 30 + 0.06*userDensity; % %
packetLoss = 0.5 + 0.005*userDensity; % %
%% Figure 1: User Density vs Latency
```

```matlab
figure;
plot(userDensity, latency,'-o','LineWidth',2);
grid on;
xlabel('User Density');
ylabel('Latency (ms)');
title('5G Core Routing: User Density vs Latency');
%% Figure 2: User Density vs Throughput
figure;
plot(userDensity, throughput,'-s','LineWidth',2);
grid on;
xlabel('User Density');
ylabel('Throughput (Mbps)');
title('5G Core Routing: User Density vs Throughput');
%% Figure 3: User Density vs Resource Utilization
figure;
plot(userDensity, resourceUtilization,'-^','LineWidth',2);
grid on;
xlabel('User Density');
ylabel('Resource Utilization (%)');
title('5G Core Routing: User Density vs Resource Utilization');
%% Figure 4: User Density vs Packet Loss
figure;
plot(userDensity, packetLoss,'-d','LineWidth',2);
grid on;
xlabel('User Density');
ylabel('Packet Loss (%)');
title('5G Core Routing: User Density vs Packet Loss');
%% Figure 5: Mobility vs Throughput
throughputMobility = 1100 - 2*mobility;
figure;
plot(mobility, throughputMobility,'-*','LineWidth',2);
grid on;
xlabel('Mobility (km/h)');
ylabel('Throughput (Mbps)');
title('Mobility vs Throughput');
%% Display Results
disp('--------------------------------');
disp('5G Core Traffic Routing Results');
disp('--------------------------------');
Result = table(userDensity', latency', throughput', ...
resourceUtilization', packetLoss', ...
'VariableNames', {'UserDensity','Latency_ms','Throughput_Mbps',...
'ResourceUtilization_percent','PacketLoss_percent'});
disp(Result);
```

Outputs :

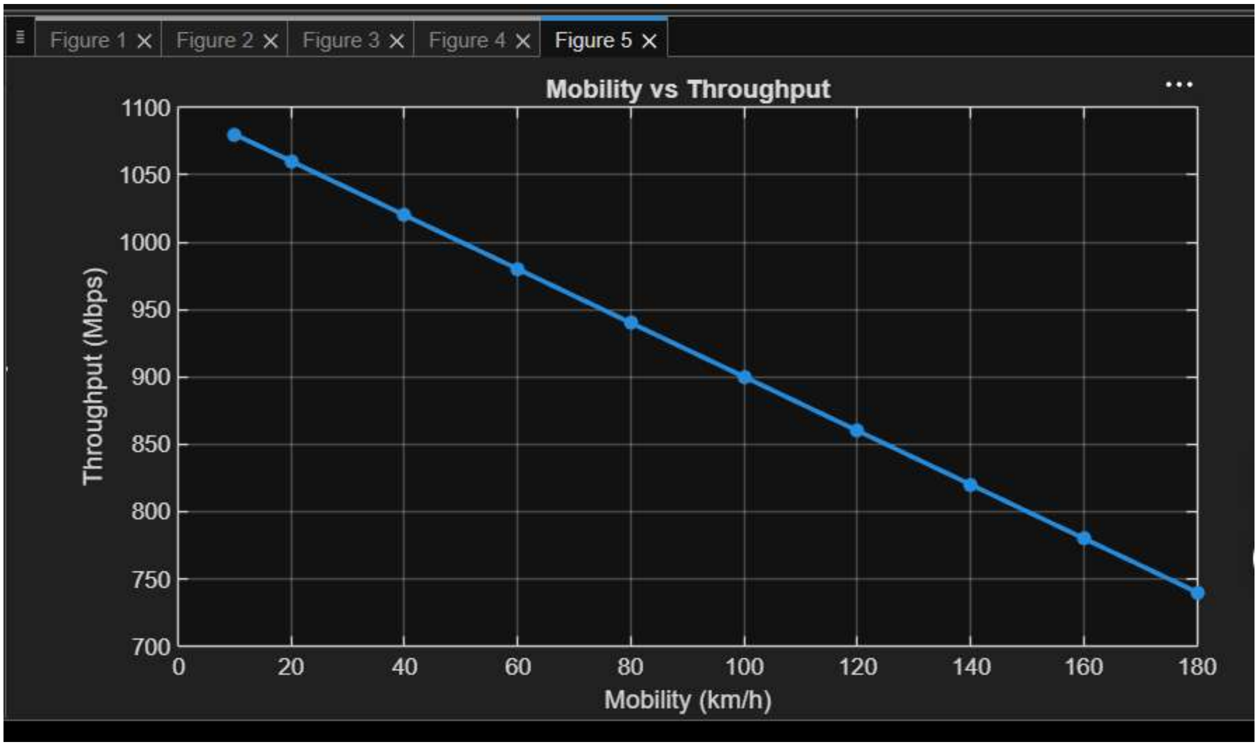
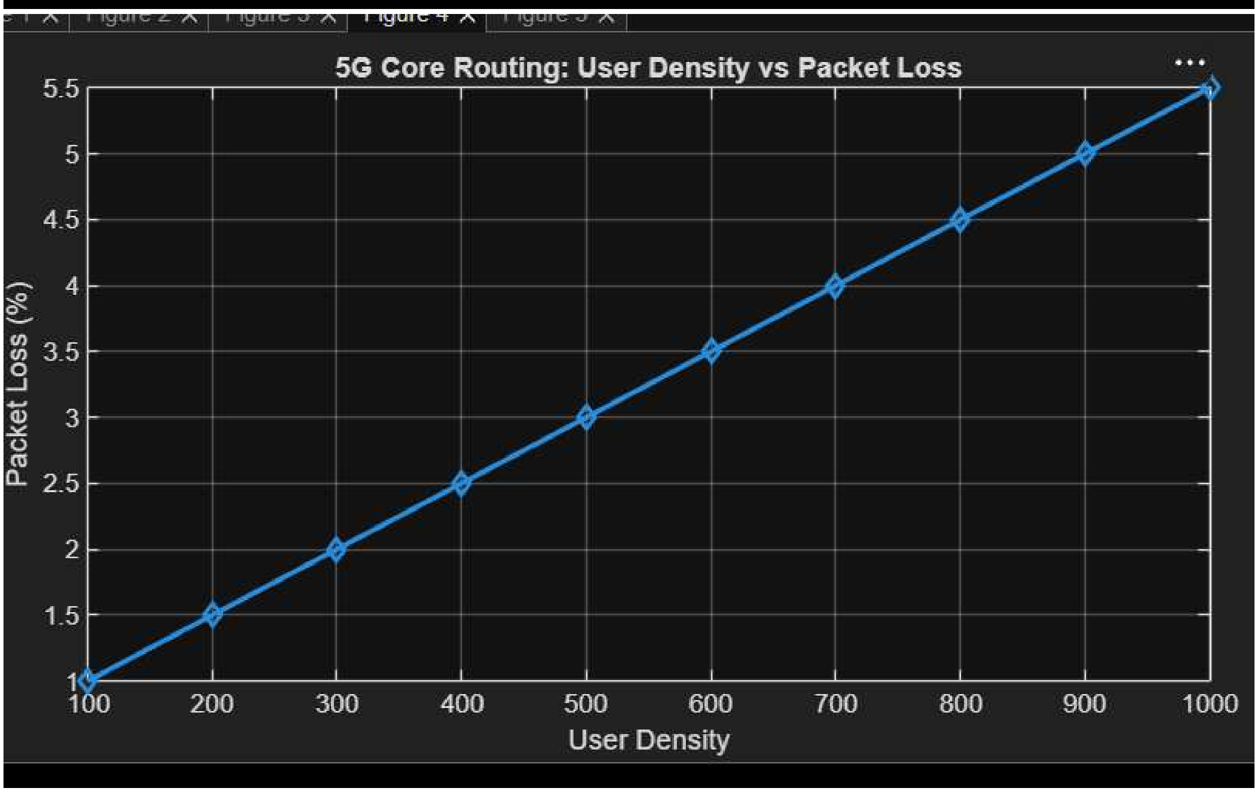
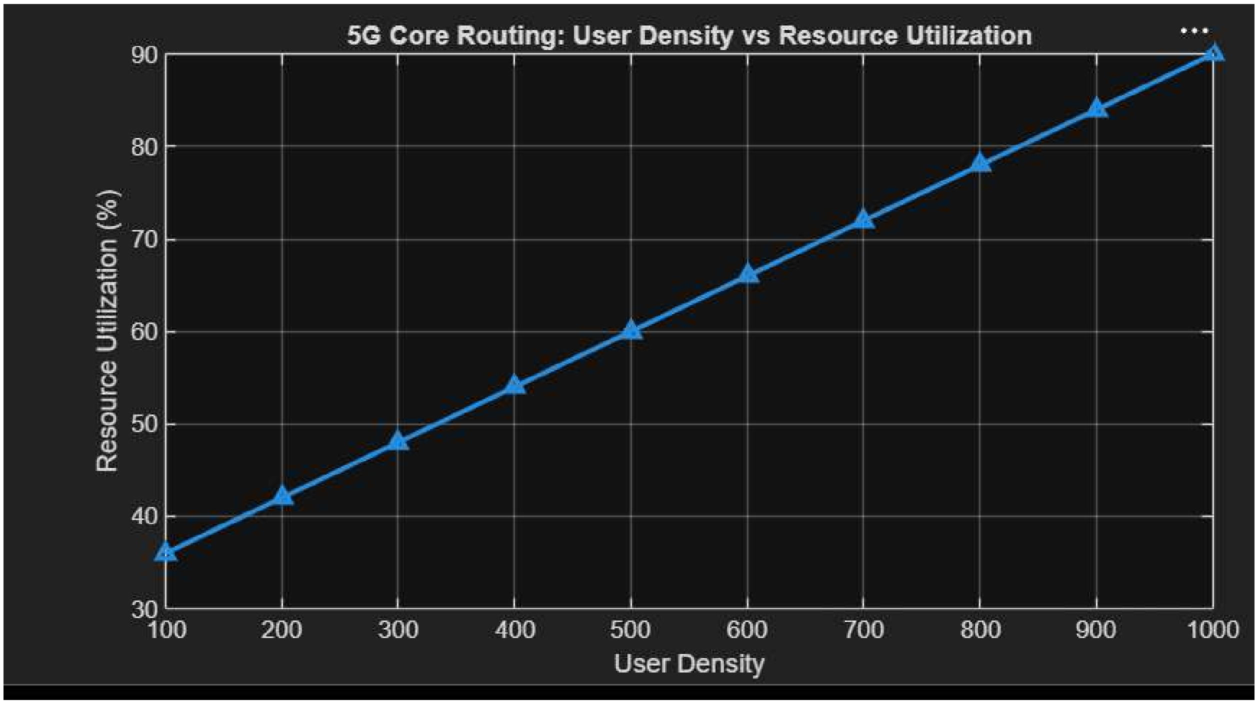


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Mobility vs Throughput
Throughput (Mbps)
Mobility (km/h)
1100
1050
1000
950
900
850
800
750
700
0
20
40
60
80
100
120
140
160
180


5G Core Routing: User Density vs Packet Loss
Packet Loss (%)
User Density
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
100
200
300
400
500
600
700
800
900
1000


5G Core Routing: User Density vs Resource Utilization
Resource Utilization (%)
30
40
50
60
70
80
90
User Density
100
200
300
400
500
600
700
800
900
1000

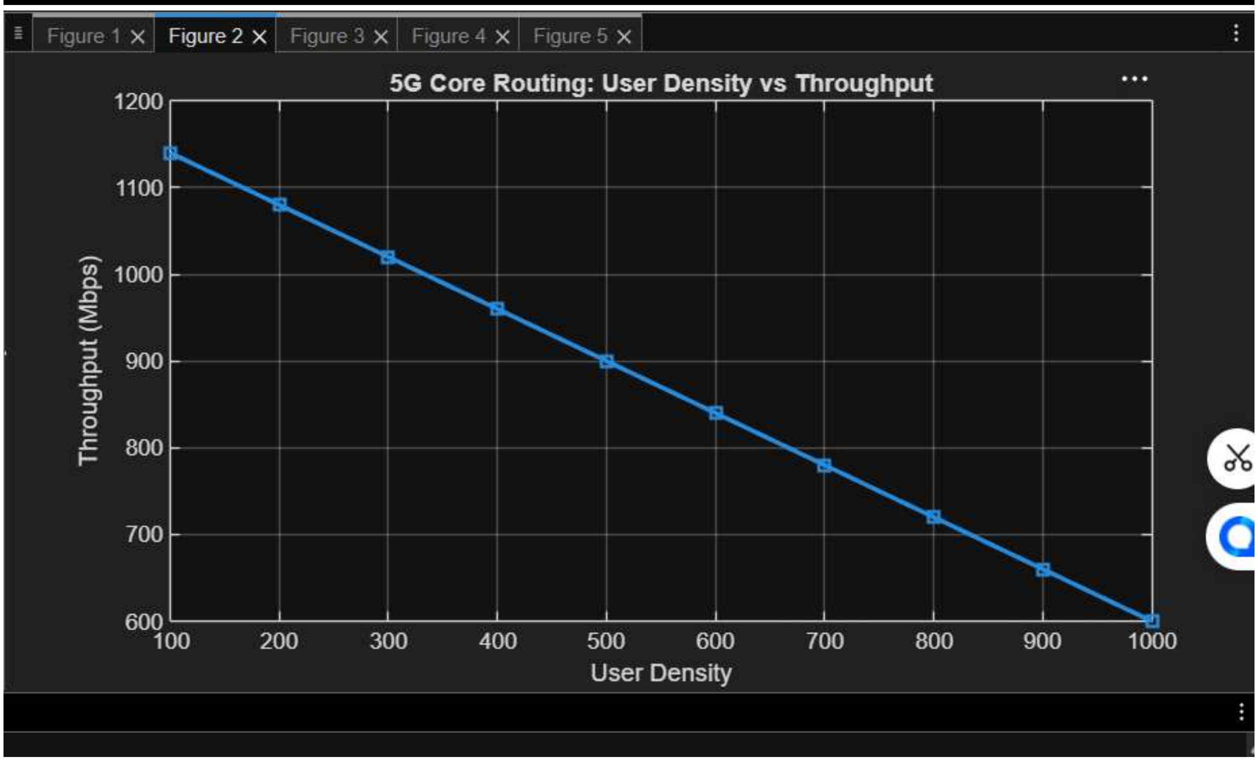


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
5G Core Routing: User Density vs Throughput
Throughput (Mbps)
600
700
800
900
1000
1100
1200
User Density
100
200
300
400
500
600
700
800
900
1000

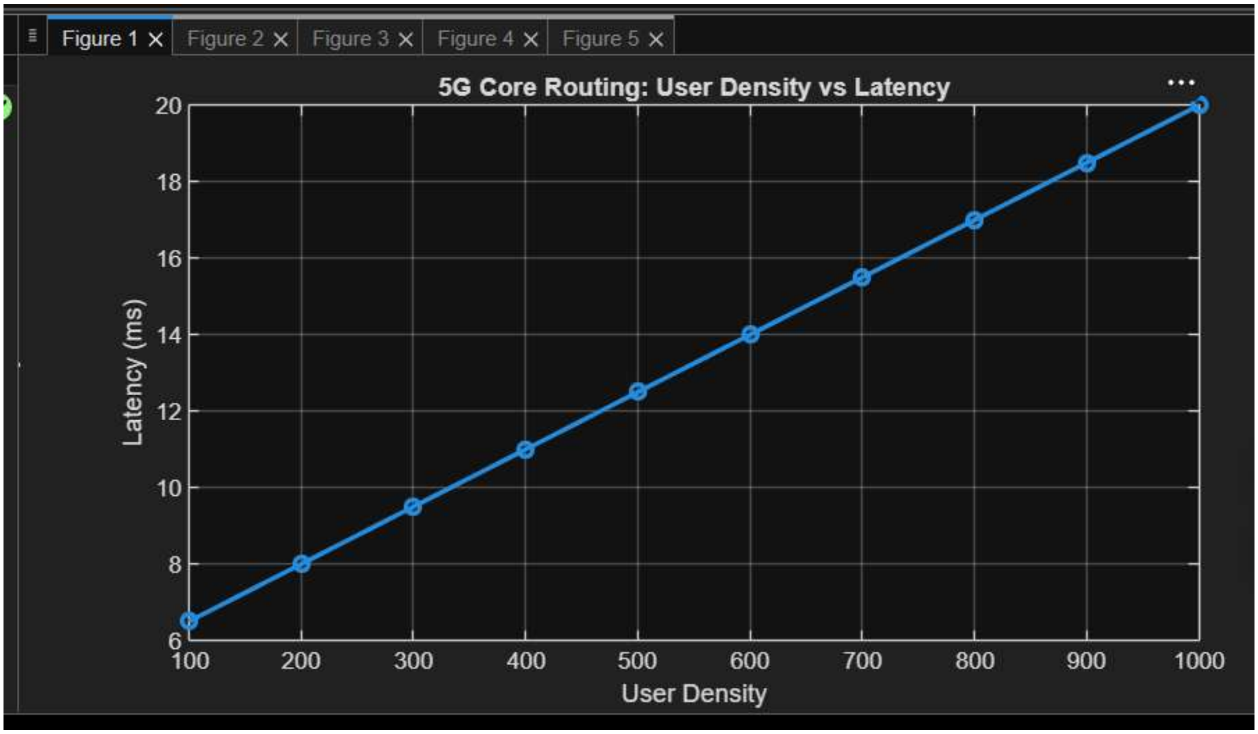


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
5G Core Routing: User Density vs Latency
Latency (ms)
20
18
16
14
12
10
8
6
100
200
300
400
500
600
700
800
900
1000
User Density